

1924–2010

BENOIT MANDELBROT

MILESTONES

NOISE PATTERNS

Accepts position at IBM analyzing noise pollution in 1958; detects set patterns within random data in 1967.

FRACTAL GEOMETRY

Coins term “fractal” for a repeating mathematical pattern; proposes theory of fractal geometry in 1975.

GAINS RECOGNITION

Receives many awards and honors, including Wolf Prize for Physics in 1993, and Japan Prize in 2003.

Polish-born French mathematician Benoît Mandelbrot founded fractal geometry, which created a new way of collecting and analyzing data.

Benoît Mandelbrot emigrated to France in 1936, and gained a PhD in mathematics from the University of Paris. He took a role at IBM in New York in 1958 and devised the concept of fractal geometry, which detects an ordered logic in seemingly random or chaotic things. This form of geometry is based on symmetry found in repeating patterns and can be applied to natural phenomena, such as coastlines or tree bark, as well as analyses of data, such as noise patterns in economics and weather patterns. It created a new way of visualizing data that has since been applied in many fields—including medicine, engineering, and cosmology—and has diverse applications such as helping to understand financial markets, predict earthquake patterns, and diagnose disease.

The Mandelbrot set is a fractal plotted using a group of complex numbers. When rendered on computers, it reveals an infinitely repeating shape and pattern.

